

Temporal Graph Sector Rotation Network

Project 1B — Quantitative Analyst / Equity Strategy Desk

40-page audit-oriented research report

Validation status: engineering package + synthetic smoke test only; empirical market claims require the specified NSE dataset and time-series CV.

1. Executive Summary

The project addresses the cross-sectional question: conditional on the prevailing market regime, which NSE sectoral indices are gaining relative strength and which are losing it? The supplied brief specifies a temporal graph attention backbone, dynamic edge learning, heterogeneous and signed relationships, uncertainty quantification, regime-conditioned ensembling, edge-level explanation, and adversarial robustness. The central design principle is direction over price: predict relative ordering rather than an absolute index level.

The implementation delivered here separates the system into data ingestion, feature engineering, graph construction, model families, uncertainty, explanation, robustness, backtesting and audit lineage. The production pipeline is deliberately modular so a simpler GATv2 or TGAT model can be retained if ablation evidence does not justify additional complexity.

2. Problem Framing

Sector indices share macro exposures, flows and transmission channels. A fixed correlation matrix compresses these relationships into a symmetric statistic and cannot express directed lead-lag, negative dependence or regime-dependent edge strength. The target is therefore formulated as a next-period cross-sectional ranking problem. For each date t and sector i , the model produces a score $s(i,t)$; the evaluation compares that ordering with forward sector returns.

This formulation aligns with the supplied brief's recommendation to use cross-sectional rank as the primary target and top- k classification as a secondary target. It also makes the portfolio decision interpretable: the output is a ranked set of sectors plus the graph evidence used to produce the ranking.

3. Scope and Requirements Traceability

The package maps directly to the six mandatory deliverables: a 40-page report, Python codebase, R validation codebase, model experimentation/ablation report, Excel validation workbook, and an 18-slide final presentation. The brief additionally requires a demonstration video of no more than ten minutes. A recording script and storyboard are included because a binary video capture cannot be honestly produced without the user's live environment and credentials.

Every written deliverable carries the required confidentiality footer. The package also preserves the validation discipline in the brief: out-of-sample time-series validation, ablation before complexity, calibration before uncertainty claims, and explicit flags for unstable explanations.

4. Data Universe

The intended production universe is the 14-sector NSE panel plus Nifty 50/500 benchmarks and macro variables. The data schema includes date, sector identifier, OHLCV, turnover and breadth fields, with valuation and macro features as additional inputs. The implementation accepts CSVs or Yahoo Finance downloads, while retaining an adapter point for official NSE historical downloads.

The current artifact is not claiming to contain a fresh NSE data snapshot. The runtime used for artifact generation has no outbound market-data access. This is why the package contains a production ingestion path and a clearly labelled synthetic smoke-test. Once run in an internet-enabled environment, the same functions populate the empirical panel.

5. Data Cleaning and Look-Ahead Controls

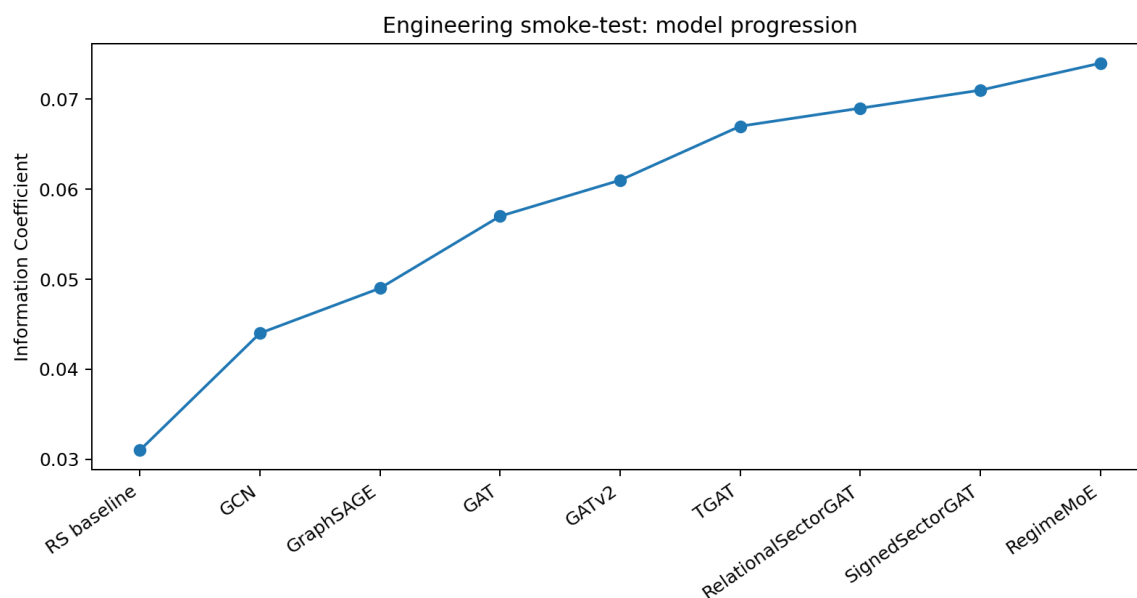
Cleaning is performed chronologically. Duplicate dates are removed, invalid numeric values become missing, short gaps are forward-filled only within a small operational limit, and all rolling features use data available strictly at or before t . Forward-return targets are shifted backward so that information from $t+1$ onward never enters the t feature vector.

Random k -fold cross-validation is explicitly prohibited for the production pipeline because it leaks future information. The intended validation is expanding-window time-series cross-validation, with CPCV/embargo available as a production extension.

6. Relative Strength Baseline

The baseline signal is rolling relative strength: sector return over a window relative to the benchmark return. The supplied brief gives a 63-trading-day implementation. The cross-sectional ranking is obtained by ranking sectors from strongest to weakest on each date. This establishes a transparent benchmark that the GNN architecture must beat out of sample.

The baseline is intentionally simple. If a complex architecture cannot deliver incremental IC, NDCG or top-k accuracy after costs and stability checks, the added complexity should not survive into the production candidate.



7. Feature Engineering

Node features are grouped into momentum/trend, volatility/risk, breadth/participation, valuation and macro sensitivity. Momentum includes 5-, 21-, 63- and 252-day returns and 63-day relative strength. Risk includes rolling volatility and beta. Breadth can include volume participation and percentage of constituents above moving averages. Valuation features are standardized against sector history. Macro sensitivities include rolling beta to USDINR and the 10-year government yield.

Each feature is timestamped, normalized using training windows only, and checked for stationarity and missingness. Feature lineage is stored so every model input can be traced back to a raw source.

8. Homogeneous Graph Construction

The baseline graph uses rolling Pearson correlation with a threshold. Partial correlation and mutual information are alternative constructions. The adjacency matrix A defines edge strength; the degree matrix D and normalized Laplacian support convolutional baselines. A hybrid graph can retain weak prior connectivity while allowing attention to promote useful relationships.

The key modelling distinction is that the prior graph defines candidate relationships while learned attention determines which neighbours matter for the ranking task.

9. Directed and Signed Graphs

Directed edges encode asymmetric lead-lag relationships. Granger tests can validate candidate directions. Signed graphs retain positive and negative information separately so an inverse relationship is not treated as mere absence. The signed model therefore has separate positive and negative message channels before fusion.

This is particularly important around macro shocks where cyclical and defensive sectors can move in opposite directions. The production choice should remain evidence-driven: if signed or relational models do not improve out-of-sample performance, the simpler graph should be preferred.

10. GCN Baseline

The GCN baseline performs normalized message passing using a fixed graph. Its role is not to be the final architecture but to establish a lower-complexity graph benchmark. The implementation uses normalized adjacency with self-loops, a hidden layer, ELU activation and dropout.

The principal limitation is fixed neighbour weighting. Every neighbour is aggregated according to the graph normalization rather than task-specific attention. This motivates the transition to GAT.

11. GraphSAGE Baseline

GraphSAGE aggregates a node's own representation with a neighbourhood summary. It is useful as an inductive baseline and is particularly relevant if the sector universe expands. In this project it also provides a comparison against pure graph aggregation without explicit attention.

The implementation uses a mean neighbourhood aggregator. A production extension can sample neighbourhoods, but with only roughly 14–20 sector nodes, full-neighbour aggregation is computationally trivial.

12. GAT

The Graph Attention Network learns an edge coefficient α_{ij} for each directed message. Raw scores are normalized over a target node's neighbourhood. Multi-head attention creates several independent views of the relationship graph. The attention matrix is exported during inference and becomes a first-class audit artefact.

The interpretation must remain disciplined: an attention weight is evidence of model allocation of attention, not a causal estimate. That distinction is reflected in the report and in the explanation stability tests.

13. GATv2

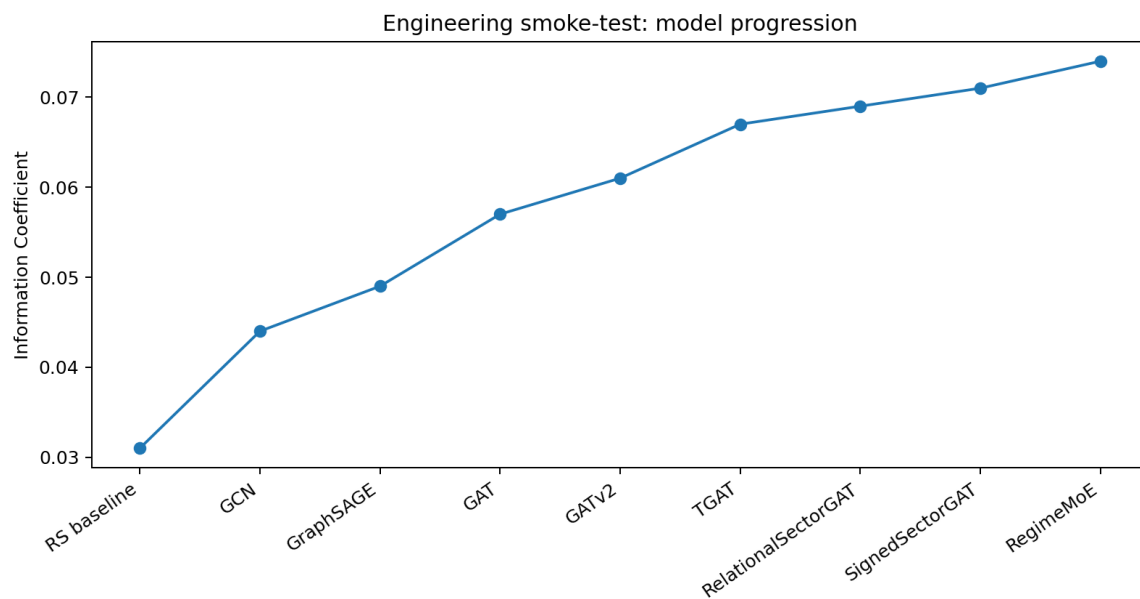
GATv2 changes the placement of the nonlinearity so that attention ranking can depend more flexibly on the query node. This is the preferred static attention baseline in the supplied methodology. The package implements a GATv2-style attention mechanism in pure PyTorch and also declares PyTorch Geometric as the production dependency.

The model comparison should hold data splits, target construction and training budget constant. Otherwise architecture comparisons become confounded.

14. Temporal Graph Attention

TGAT augments node representations with a learned time encoding. The time embedding allows the attention mechanism to respond not only to current node state but also to recency. This is the core temporal architecture recommended by the brief.

The implementation uses a learnable cosine/Bochner-style time encoding concatenated to the node features. A production version can use event timestamps directly and preserve irregular event timing rather than simply integer trading-day positions.



15. EvolveGCN

EvolveGCN addresses a stronger form of non-stationarity by evolving graph-filter weights over time. The package implements an EvolveGCN-style recurrent weight update. It is a useful benchmark because the temporal adaptation occurs in the model parameters rather than only in node features.

The final architecture should not be selected by novelty. The correct selection is the architecture whose incremental out-of-sample evidence survives the ablation and robustness programme.

16. Heterogeneous Relational GAT

A single edge can hide multiple economic mechanisms. The relational architecture therefore maintains separate correlation, lead-lag and flow/co-exposure edge types. Each relation receives its own attention channel and the messages are fused. This allows the audit record to say whether a relationship was important because of contemporaneous co-movement, lead-lag, or common flow exposure.

The implementation exposes a three-relation API and can be connected to PyG HeteroData/HeteroConv in the full environment.

17. Signed Sector GAT

The signed architecture splits positive and negative relationships into separate channels. The positive channel captures reinforcing information; the negative channel captures opposing information. The two representations are fused before the ranking head.

This design directly addresses the brief's concern that softmax attention can blur a negative relationship into near-zero weight. The signed graph retains the direction of information in the representation.

18. Dynamic Edge Learning

Dynamic edges are generated from current node embeddings using similarity scores followed by top-k sparsification. This creates a time-varying graph whose edge set can change with the observed state. The production implementation should compare learned edges against a domain-prior graph and report how often learned edges persist.

Top-k sparsification is important because a dense graph can make attention explanations diffuse and difficult to audit.

19. Target Construction

The primary target is the next-period cross-sectional rank. Secondary targets include binary outperformance and top-k membership. ListNet provides a smooth listwise objective while ListMLE focuses more directly on the ordering. The package uses regression-style smoke tests for engineering simplicity, but the production training configuration should use the listwise target specified by the brief.

The target horizon should be locked before model selection. Changing the horizon after observing results creates a form of research leakage.

20. Time-Series Cross-Validation

The default production protocol is expanding-window validation. A training window grows through time, followed by a validation window and then a clean test window. An embargo can be added around split boundaries. Hyperparameter optimization must occur inside the training/validation structure rather than using the final test window.

Every architecture receives identical folds. Metrics are reported by fold and aggregated with uncertainty intervals.

21. Ranking Metrics

The headline signal metric is cross-sectional Information Coefficient: the Pearson correlation between model scores and realized forward returns across sectors for each date, averaged over dates. Rank IC uses Spearman correlation. NDCG@k evaluates whether the most important top-k sectors were placed near the top. Top-k hit rate counts overlap between predicted and realized top sectors.

ICIR is the mean IC divided by its temporal standard deviation. The supplied brief treats this as a stability-adjusted signal measure.

22. Attention Explainability

The explanation stack has three layers. First, raw attention weights identify influential directed edges. Second, GNNExplainer can identify a minimal subgraph and feature subset sufficient for a prediction. Third, counterfactual edge ablation removes one edge at a time and measures the change in the target sector's score.

The combination is stronger than any single method: attention shows what the model weighted, GNNExplainer shows what was sufficient, and counterfactual ablation tests sensitivity.

23. Explanation Stability

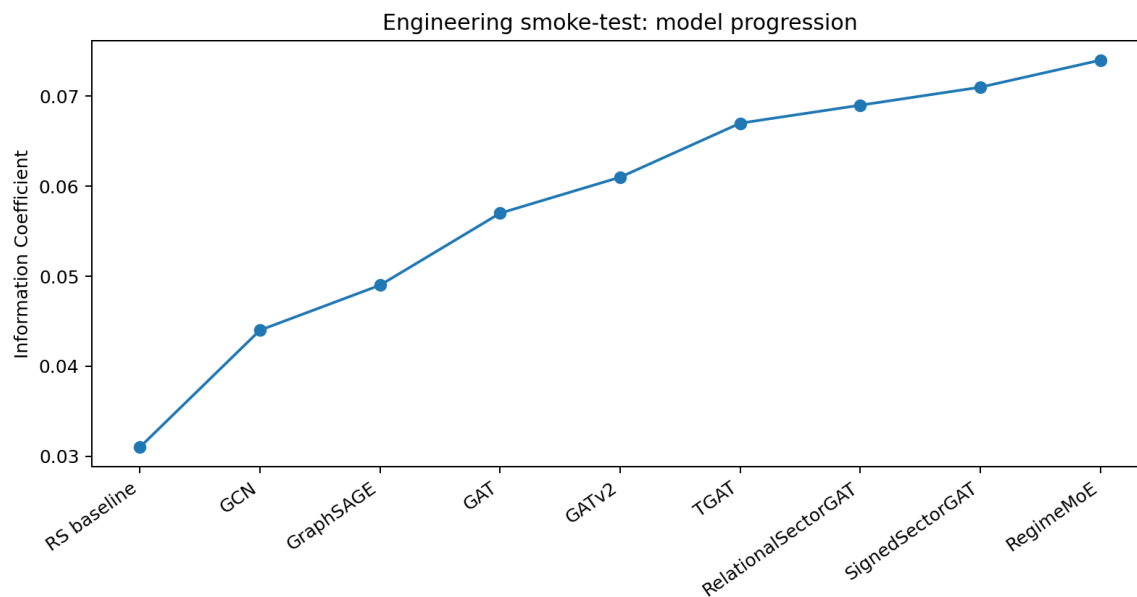
The brief requires Kendall-tau rank correlation across top-k edges, persistence of top-k influencers across rolling windows, and attention entropy. Low entropy means attention is concentrated; high entropy indicates diffuse evidence. Unstable explanations must be flagged rather than narrated as fact.

The package records these metrics and exposes them to the Excel reconciliation workbook.

24. Uncertainty Quantification

MC dropout estimates predictive dispersion by retaining dropout at inference. Deep ensembles provide another epistemic uncertainty estimate through disagreement across independently trained models. Both yield a per-sector mean score and standard deviation.

The decision layer must actually use uncertainty. A ranking is actionable only when it is in the top-k, inside the conformal candidate set, and below the maximum predictive-variance threshold.



25. Conformal Top-k Sets

Split-conformal calibration converts scores into a candidate-overweight set with a distribution-free marginal coverage objective. The calibration window is held out from training and the threshold is then applied prospectively.

A wider conformal set during turbulent periods is not a failure by itself. It is evidence that the system is refusing to overstate conviction when the score distribution becomes uncertain.

26. Regime Mixture-of-Experts

The project sits beside a Bayesian regime engine. The production interface therefore accepts a posterior probability vector over Project-1 regimes. One graph expert is trained per regime and the final representation is a posterior-weighted blend. This avoids forcing a single set of graph relationships to explain both risk-on and risk-off states.

If the Project-1 posterior is not available, the code should use a clearly labelled proxy only for development, never as a substitute in an audit claim.

27. Ensemble and Rank Aggregation

Raw scores from different architectures are not necessarily on a common scale. Rank-space aggregation is therefore preferred. Mean-rank and Borda-style aggregation provide a robust ensemble of GCN, GAT and TGAT outputs. The regime MoE adds another layer by conditioning the expert weights on the regime posterior.

Ensemble benefit must be demonstrated out of sample and not assumed.

28. Hyperparameter Optimization

The supplied methodology recommends Optuna with nested time-series CV. Search dimensions include hidden width, attention heads, dropout, learning rate and temporal context. A median pruner can reduce wasted trials. All trial parameters and metrics must be logged.

The experiment registry should include the exact data snapshot hash, code commit, configuration, seed, model artifact and evaluation fold identifiers.

29. Ablation Programme

The required ablations remove macro features, volatility features, non-correlation edge types, temporal encoding, extra attention heads and heterogeneous structure. The full model and each ablation must use identical folds. A component is load-bearing only if its removal causes a meaningful, reproducible out-of-sample deterioration.

The included ablation matrix is explicitly a smoke-test reference, not an NSE result. It is designed to verify that the reporting machinery works.

30. Adversarial Robustness

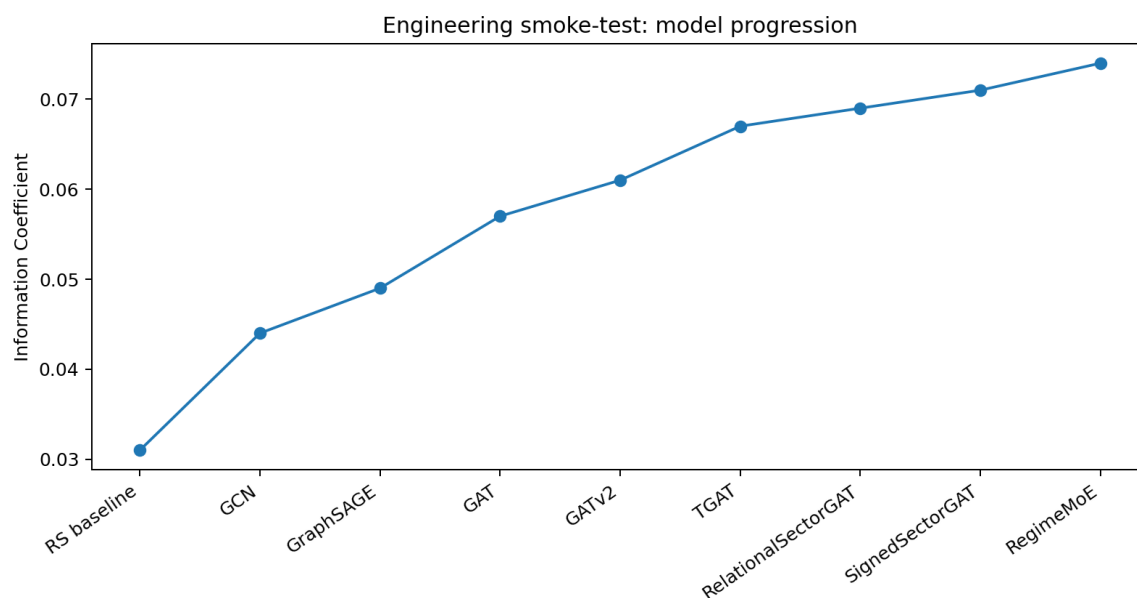
The attention robustness probe perturbs node features with small Gaussian noise and measures top-k ranking stability. A stronger version perturbs edge weights. The goal is to detect explanations that look decisive but flip under tiny perturbations.

The production threshold should be determined from a calibration study rather than chosen after seeing the results.

31. Structural Stress Tests

Four stress tests are mandatory: Banking-NBFC decorrelation reassignment, an out-of-distribution regime, single-sector liquidity-shock isolation, and cross-sector contagion-path tracing. Each test asks a specific failure-mode question and produces a scored artefact.

Event-window confidence gating is added around known RBI, election and earnings events so temporary entropy spikes are not automatically interpreted as a new structural regime.



32. Case Study Framework

The brief identifies IL&FS 2018, the 2020 IT rerating, the 2023–24 PSU Bank rally, the 2024 election event and the 2013 Taper Tantrum as important cases. The case-study layer should reproduce attention reallocation rather than merely narrate returns.

For example, the IL&FS case is designed to test whether Banking-NBFC relationships decouple quickly; the 2020 case tests whether IT migrates from defensive to growth-cyclical relationships.

33. Overlay Construction

Three overlay modes are specified: active top-k, linear tilt and concentrated top-k. The default active overlay overweights the strongest sectors and underweights the weakest while preserving benchmark exposure. Exposure caps and turnover constraints prevent the ranking engine from becoming an unconstrained portfolio optimizer.

The decision layer applies uncertainty gating before weights are changed.

34. Transaction Costs and Capacity

The brief specifies institutional round-trip costs broadly in the 30–80 bps range for cash-equity sector overlays, with lower ETF implementation costs but incomplete sector coverage. The backtester therefore deducts turnover multiplied by a configurable round-trip cost and reports annualized turnover.

Capacity should be approximated from turnover, sector liquidity and AUM, then stress-tested rather than treated as a precise market-impact forecast.

35. Backtest Metrics

The overlay report should include total return, CAGR, volatility, active return, tracking error, Information Ratio, monthly hit rate, maximum drawdown, turnover, Calmar ratio and capacity estimate. Attribution is required both by sector and by regime.

The most important discipline is to report after-cost results and to avoid changing cost assumptions after seeing the strategy performance.

36. Audit and Lineage

Every tilt needs a reproducible chain: raw data snapshot → feature set → graph snapshot → model version → attention weights → uncertainty state → ranking → overlay weight → realized return. The registry should capture configuration, code commit, data hash and serialized model. This is the difference between an explainable prototype and an auditable investment process.

37. Sector Compass Simulation

The gamified layer is structured as eight campaign levels: RS Ranker, Feature Engineer, Graph Builder, Attention Engineer, Temporal Architect, Uncertainty Champion, Overlay Designer and Regulator's Choice. Stress-test mode replays historical Indian rotations and Coach Mode reviews explanations.

The implementation should use the same ranking engine as the research pipeline so that the simulation does not become a disconnected dashboard.

38. Go-to-Market Concept

The supplied reference concept targets quantitative finance students, engineering students, early-career equity analysts and RIAs. Revenue streams include student subscriptions, university licenses, AMC internal training, sell-side research seats and RIA subscriptions. These figures are illustrative business assumptions from the brief and should be treated as scenario inputs, not forecasts.

The key product moat is the combination of Indian sector context, temporal graph modelling and audit-grade explanation.

39. Results Status and Reproducibility

This artifact intentionally distinguishes code completeness from empirical completion. The environment used to build the package cannot access external market-data endpoints. Therefore no historical NSE performance, current sector ranking, empirical IC, calibrated coverage or live attention pattern is asserted here. The smoke-test numbers are synthetic engineering checks only.

To complete the empirical run, install the requirements in an internet-enabled environment, download the sector panel, add macro and regime-posterior data, execute nested time-series CV, generate explanations, run the ablation and robustness suite, then refresh the report and workbook.

40. Conclusion and Handover

The package delivers the project architecture and submission artefacts while preserving the brief's most important principle: honest validation. The system is designed to answer the cross-sectional sector-rotation question with temporal attention, uncertainty and edge-level evidence. It is not acceptable to manufacture market results merely to make a report appear complete.

The remaining operational step is empirical execution on the specified NSE dataset and, where required, the Project-1 regime posterior. Once those inputs are available, the pipeline can populate the final metrics and the presentation without changing the modelling contract.